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Software requirement specification (SRS) for Inventory management system for ASTU

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Abstract

This document presents the Software Requirement Specification (SRS) for the Inventory Management System (IMS) designed for Adama Science and Technology University (ASTU). Designed by 3rd year software engineering students. The goal of the system is to modernize and streamline the management of universty assets, supplies, and materials by replacing manual and inconsistent processes with an efficient, centralized digital platform.

The proposed IMS will enable departments and administrative offices to record, track, update, and monitor inventory items in real time. It supports key functions such as item registration, stock tracking, purchase requests, approval workflows, supplier management, and automated reporting. The system aims to improve accuracy, reduce losses, support decision-making, and enhance transparency across all inventory-related activities.

This SRS outlines the functional and non-functional requirements, system features, data models, user roles, and workflows necessary for the successful development of the IMS. The document serves as a foundation for designers, developers, testers, and stakeholders involved in the project. Although developed for ASTU, the system can be customized and implemented for any organization that needs effective and reliable inventory management.

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CHAPTER ONE: INTRODUCTION

1.1 Background of the project

Inventory is one of the most important resources for any organization because it directly affects service delivery, cost control, and operational efficiency. Many institutions, including universities, still rely on manual or semi-manual inventory handling such as paper records, Excel sheets, or departmental logs. These methods often cause **delays**, **errors**, **stock shortages**, and **difficulty in tracking materials.**

To solve these issues, modern organizations are shifting to computerized Inventory Management Systems (IMS). A computerized IMS helps to **record items**, **monitor stock levels**, **manage purchase requests, track item movement**, **generate reports**, and **provide real-time information for decision making.** Therefore, this project aims to analyze, design, and document the requirements for a new IMS that improves accuracy, transparency, and efficiency of warehouse and store management.

1.2 statement of the Problem

The current inventory handling process faces several challenges such as:

- Inaccurate records due to manual entry error.
- Difficulty in tracking items issued, returned, or transferred due to loss of document.
- Slow purchase request and approval process.
- Lack of timely reports for management.
- Misplacement or duplication of items.
- Limited transparency and accountability.

These issues negatively affect the overall performance of the store, causing delays, unnecessary costs, and stock shortages.

1.3 Purpose of the Project

The project purpose revolves around creating an inventory management system for warehouse in ASTU. The project details include **stock information**, **managing stock**, and **ordering stock and review the stock as a report.**

This system aims to:

- ✓ Eliminate human errors associated with manual record-keeping.
- ✓ Improve workflow efficiency, allowing staff to focus on customer engagement rather than administrative burdens.
- ✓ Optimize decision-making with real-time tracking and inventory reports.

1.4 Team composition

This project is carried out by a group of 5 students working collaboratively. In that time each students have their own task and contribute in **requirement collection, documentation**, **analysis**, **design**, and **review** according to assign roles.

1.5 Objectives of the project

1.5.1 General objective

To prepare a complete SRS document for a computerized Inventory Management System that improves the existing manual store operations.

1.5.2 Specific objective

- ✓ Identifying the problems in current manual system.
- ✓ To gather system and user requirement.
- ✓ To specify functional and non-functional requirements.
- ✓ To design system models (use case diagram, class diagram, sequence diagram,...).
- ✓ To propose feasible and efficient system architecture.
- ✓ To prepare detailed documentation for future system development.

1.6 Feasibility study

1.6.1 Technical Feasibility

The system will be developed using widely available tools such as relational databases, web technologies (like HTML, CSS), and standard programming languages (java/servlet). The required hardware and software are accessible and affordable.

1.6.2 Operational Feasibility

Store workers, managers, and administrators can easily operate the system with minimal training. The system workflow fits the existing organizational processes. Additionally, stakeholders have shown positive interest and willingness to adopt the system, which increases the likelihood of successful implementation.

1.6.3 Economic Feasibility

The project requires minimal cost, mainly for development tools and training. It reduces long-term operational costs by minimizing errors, losses, and extra administrative work. The cost of development and maintenance is reasonable compared to the benefits gained, such as improved efficiency, reduced errors, better inventory control, and enhanced transparency.

1.7 Scope and limitations

1.7.1 Scope

The Inventory Management System (IMS) for Adama Science and Technology University (ASTU) is designed to streamline and centralize the management of institutional assets and materials across all departments and administrative units. The system will provide a digital platform for recording, tracking, updating, and reporting inventory items such as computers, laboratory equipment, and office supplies.

The inventory management System encompasses several essential functions designed to enhance store operation and improve efficiency.

They are:

- Real-time tracking of inventory items across departments
- Request and approval workflows for asset allocation and procurement
- Automated report generation for audits and decision-making
- Stock level monitoring and alerts for restocking
- Role-based access control for warehouse manager, department heads, and staff.

1.7.2 Limitation

Despite its comprehensive design, the IMS will have the following constraints:

- The system will require stable internet connectivity for real-time updates and multi-user access.
- Integration with existing legacy systems at ASTU may be limited due to compatibility issues.
- Mobile access will be limited to web-responsive interfaces.
- Automated synchronization with external procurement or financial systems is not supported in the first version.
- The system does not include predictive analytics or AI-based forecasting for inventory trends.
- User training and change management are outside the scope of the software but are essential for successful adoption.

1.8 Significance of the project

The development of the IMS plays a crucial role in enhancing stock operations and improving staff experiences. By transitioning from manual processes to digital system.

The significance of this project multifaceted:-

- Optimization resources and cost reduction
- Enhanced operational efficiency
- Real time data for decision making

Beneficiaries from this project incude:

- University administration
- Store keepers
- Each Staff/department members

1.9 Methodology

1.9.1 Data Collection Methodology

Data is gathered through:-

- Interviews with warehouse managers and staff members.
- Observation of existing manual system.
- Document analysis of stock records in the store.

1.9.2 System development methodology

Agile Development Methodology

To ensure flexibility and continuous improvement, the inventory Management System will be developed using an Agile methodology. Even if the other methodology is applicable for this project we chose this approach to prioritizes iterative development, allowing for regular feedback and adjustments throughout the project lifecycle.

1.10 Development tools

Tools used in this SRS documentation include:

- Microsoft Word for documentation
- Draw.io for diagrams
- Telegram group for team communication and information exchange.
- Not pad in interview to record interviewers answer.

1.11 Required resources and cost estimation

Resources for development and deployment

The system need hardware and software resources like computer, server, operating system, programming language, database management system, design tools, documentation tools and it also need human resource that can able to build the system and interact with them.

Cost estimation

The cost is minimal due to the several reason we can use easily our personal computer to build the system and the staff member also have their own computer. So, they can easily use that for our web. And also there are internet connection in some place around the ASTU compound so it is not necessarily add additional cost for this purpose.

1.12 Task scheduling and work plan

The project activities were divided into phases such as requirement gathering, analysis of the existing system, documentation, system modeling, and review.

Each task was assigned a specific duration and responsibility among team members. Proper scheduling helped the team manage time effectively, avoid delays, and ensure that all project objectives were met. Regular progress reviews were conducted to monitor task completion and address challenges promptly.

1.13 Testing procedure

To ensure system reliability and performance, the following testing phases will be conducted:

Unit Testing Individual components and modules will be tested separately to verify their functionality and identify potential errors early in the development process.

Integration Testing Once individual modules are validated, integration testing will assess interactions between different components to ensure seamless operation.

System Testing This involves testing the complete, integrated system to evaluate its compliance with the specified functional and non-functional requirements.

User Acceptance Testing (UAT) End-users will test the system's functionality against business requirements, providing feedback to refine usability and performance before final deployment.

1.14 Definition of Terms and Acronyms

This section provide some abbreviation that are used in this documentation with their full description to improve the understanding to ensure that all readers, including technical and non-technical users to have a common understanding.

List of term	Definition
Admin	System administrator responsible for managing the system.
User	Authorized person who uses the system
Stock	Items stored in the warehouse
Model 19	A formal form that currently used to request issuance item
Model 22	A formal form that is confirmation of the approval item

Acronyms	***Full form***
IMS	Inventory Management System
SRS	Software Requirements Specification
SKU	Stock Keeping Unit
MOD19	Model 19
MOD22	Model 22
UC	Use case
DH	Department Head

1.15 Document Organization

This section describes how the SRS document is organized and provides a brief overview of each chapter. It helps readers easily locate information and understand the structure of the document.

The document is organized into several chapters.

Chapter One

Introduces the project and explains the background, problem statement, objectives, feasibility, scope, and methodology.

Chapter Two

Describes the existing system, including its functions, users, and limitations.

Chapter Three

Presents the proposed system, including functional and non-functional requirements and system models.

Chapter Four

Focuses on system design, architecture, database design, and user interface.

Chapter five

It include the testing strategy for the system the rest chapters are about the deployment and installation of the system focused on how to use? How to make rely in use?

The document also includes appendices that provide supporting information such as diagrams, acronyms, and additional references.

Chapter Two: Overall Description of the Existing System

2.1 Background and Description of the Existing System

The IMS at **Adama Science and Technology University (ASTU)** currently operates **mainly using a manual, paper-based approach**. Although a web-based inventory system was previously developed, it is **not actively used** by storekeepers and department staff because it is new system and currently they are not adapt the system. As a result, most inventory-related activities such as requesting, approving, issuing, and recording items are handled manually.

Types of inventory item

There are 3 types of item. That are:-

- **Non- Consumable** (Fixed asset) are long term tangible assets. Like computer,machin…
- **Consumable** (Variable asset) are assets that can fluctuate in quality or value base on usage or demand. Like; -marker, cleaning material
- **Fixed-variable** asset are asset have variable usage and less than 10,000 birr cost.(optional generally there are two type)

According to their type there are also two stores to store these types of inventory item that are fixed asset store and variable asset store. And without this two main store there is also satellite store that store department equipment's.

Current manual workflow

The current inventory workflow is fully manual and follows these steps:

Step 1: *Request Initiation*

An instructor or lecturer writes a **formal request letter in MOD19 form** for the required item and submits it to the department head.

Step 2: *Department Head Verification*

The department head checks item availability using a **manually prepared list** provided by the storekeeper.

Step 3: *Approval or Rejection*

- If the item is **available**, the request is **approved**.
- If the item is **not available**, the request is **rejected**.

Step 4: *Item Issuance*

- **Non-consumable items** are collected from the main inventory store.
- **Consumable items** are issued from department stores.

Step 5: *Manual Record Keeping*

All transactions are recorded manually in Paper logs and In the Document of the receiver.

There is **no centralized database or real-time update mechanism**.

If the receiver is withdraw from his work it can transfer to the other by using formal form.

2.2 Major Functions of the Existing System

The existing manual inventory system performs the following major functions:

- Receiving item requests from instructors and lecturers through written letters called MOD19 form.
- Verifying item availability using manually prepared item lists.
- Approving or rejecting requests by department heads.
- Issuing consumable and non-consumable items from stores.
- Recording issued items in paper-based documents such as notebooks and ledger books called **MOD22** form.

2.3 Users of the Current System

The main users (actors) involved in the current inventory system include:

- **Instructors and Lecturers** – initiate item requests.
- **Department Heads** – verify and approve or reject requests.
- **Main Storekeeper** – manages non-consumable items in the main store.
- **Department Storekeepers / Lab Assistants** – manage consumable items in department stores.
- **University Staff** – assist in inventory handling and record keeping.

2.4 Drawbacks of the Current System

The existing manual system has several limitations, identified through interviews and observation:

2.6.1 Operational Problems

- The request and approval process is slow and time-consuming
- Manual handling causes delays in item issuance
- Errors may occur during manual processing

2.6.2 Record-Keeping Problems

- Paper records can be lost or damaged.
- Inconsistent data across different stores.
- Difficulty in tracking who received which item and when.

2.6.3 Visibility and Coordination Issues

- No real-time communication between main store and department stores.
- Stock levels are not updated automatically.
- No alert for low-stock items.
- Generating reports is difficult and time-consuming.

2.6.4 System Usability Issue

Although a web-based inventory system exists:

- It is not used due to complicated user interface.
- Staff encounter usability issues during operation.
- Poor user experience led staff to return to manual methods.

2.7 Business Rules of the Existing System

The following business rules govern the current inventory operations:

- Every item request must be approved by the department head.
- Items can only be issued after written approval.
- Non-consumable items must be issued from the main store.
- Consumable items are issued from department stores.
- All issued items must be recorded manually.

2.6 Gap Analysis

Currently, the inventory management process at ASTU is mostly manual and paper-based. Although a web-based inventory system exists, it is not actively used due to poor usability and complexity. As a result, staff rely on handwritten records, letters, and manual approvals. This creates inefficiencies, errors, and delays in inventory operations.

The proposed Inventory Management System aims to address these gaps by providing a simple, user-friendly, and centralized digital solution.

Identified Gaps

1. **Process Efficiency Gap**
   - ✓ *Existing System:* Inventory requests, approvals, and item issuance are handled manually using paper documents, which is slow and time-consuming.

- ✓ *Proposed System:* Requests, approvals, and issuance will be processed electronically, reducing delays and improving workflow efficiency.

2. **Record Management Gap**
   - ✓ *Existing System:* Records are kept in notebooks, paper logs which can be lost, damaged, or inconsistent.
   - ✓ *Proposed System:* All inventory data will be stored in a centralized database, ensuring data accuracy, consistency, and easy retrieval.
3. **Visibility and Communication Gap**
   - ✓ *Existing System:* There is no real-time communication between the main store, department stores, and offices. Stock status is not updated instantly.
   - ✓ *Proposed System:* The system will provide real-time inventory updates and shared access among authorized users across departments.
4. **Tracking and Accountability Gap**
   - ✓ *Existing System:* It is difficult to track who received which item and when, especially for non-consumable assets.
   - ✓ *Proposed System:* The system will track item issuance history, user details, and asset location, improving accountability.
5. **Reporting Gap**
   - ✓ *Existing System:* Generating reports such as monthly summaries or stock status requires manual counting and checking of records.
   - ✓ *Proposed System:* The system will automatically generate reports, including stock levels, issued items, and low-stock alerts.
6. **Usability Gap**
   - ✓ *Existing System:* The previously developed web system is not used because the interface is complicated and not user-friendly.
   - ✓ *Proposed System:* The new system will focus on simplicity, ease of use, and user-centered design to encourage adoption by staff.

Implementing the proposed system will significantly improve inventory control, efficiency, and decision-making within the university.

CHAPTER THREE: PROPOSED SYSTEM REQUIREMENTS

3.1 Overview of the Proposed System

The proposed system is a **Web-Based IMS** for **ASTU**. The system is designed to replace the existing **manual, paper-based inventory process** and address the usability problems of the previously developed but unused web system.

The proposed IMS will **digitalize inventory operations**, including item requests, approvals, issuance, tracking, and reporting. It will provide a **centralized database** that connects the **main store**, **satellite stores**, and **offices/laboratories**, enabling real-time visibility and coordination.

The system will be accessible through a web browser and will support different user roles such as **Instructor, Department Head, Inventory Officer, Storekeeper, and System Administrator**.

3.2 Functional Requirements

From User Management perspective

- ✓ The system shall allow the Admin to create, update, and delete user accounts.
- ✓ The system shall assign roles and permissions based on user type.

From Authentication and Authorization perspective

- ✓ The system shall require users to log in using secure credentials (password & username).
- ✓ The system shall restrict access based on user roles.

From Inventory Item Management perspective

- ✓ The system shall allow the Admin or Inventory Officer to add, update, and remove inventory items.
- ✓ The system shall categorize items as **consumable** or **non-consumable**.
- ✓ The system shall store item details such as name, category, quantity, location, and status.

From Item Request Management perspective

- ✓ The system shall allow instructors and department heads to submit item requests online.
- ✓ The system shall record request details including item type, quantity, and reason.

From Approval Workflow perspective

- ✓ The system shall route requests to the Department Head for approval.
- ✓ The system shall allow the Inventory Officer to verify approved requests.
- ✓ The system shall notify the Storekeeper after final approval.

From Item Issuance perspective

- ✓ The system shall allow the Storekeeper to issue items after approval.
- ✓ The system shall automatically update stock quantities after issuance.

From reporting perspective

- ✓ The system shall generate reports such as issued items, available stock, and monthly summaries.
- ✓ The system shall allow reports to be viewed and printed.

From Notification perspective

- ✓ The system shall notify users about request status updates.
- ✓ The system shall alert inventory staff when stock reaches low levels.

3.3 Non-Functional Requirements

3.3.1 Organizational Requirements

- The system shall follow ASTU inventory policies and approval hierarchies.
- The system shall support accountability and transparency in inventory operations.
- The system shall be easy to learn for staff with basic computer knowledge.

3.3.2 External Requirements

- The system shall ensure data confidentiality and role-based access control.
- The system shall operate securely over the campus network.
- The system shall comply with institutional data handling rules.
- The system shall reduce paper usage and support environmentally friendly practices.

3.4 Domain and Inverse Requirements

Domain Requirements

- Inventory items must be categorized as consumable or non-consumable.
- Approved requests must exist before items are issued.
- Stock quantities must never fall below zero.

Inverse Requirements

- The system shall not allow unauthorized users to issue items.
- The system shall not allow issuance without prior approval.
- The system shall not allow deletion of inventory records without proper authorization.

3.5 Design and Implementation Requirements

- The system shall be implemented as a **web-based application**.
- The user interface shall be **simple and user-friendly**, addressing the issues of the previous system.
- The backend shall handle business logic and approval workflows.
- The system shall use a **relational database** to store inventory and transaction data.
- The system shall support future expansion and maintenance.

3.6 system models

System models help stakeholders and developers clearly understand the structure, behavior, and interaction of the proposed system before implementation.

The system model of the proposed IMS is represented using: Scenarios, Use case models, Object model, Dynamic models (sequence, activity, and state diagrams)

These models together describe **what the system does?, how it does?**, and **how different parts of the system communicate?**.

3.6.1 Scenarios

Scenario 1: *Requesting an Inventory Item*

- **Actors:** Instructor, Department Head, Inventory Officer, Storekeeper, System.

- **Outcome:** Item is issued, stock reduces, and request is archived digitally.

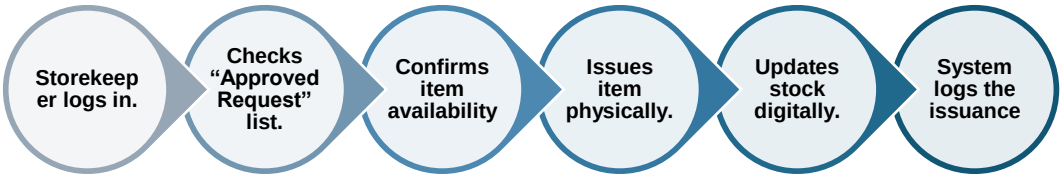
Scenario 2: *Approving an Inventory Request*

- **Actors:** Department Head, Inventory Officer

- **Outcome:** Approved requests move to issuance queue.

Scenario 3: *Issuing an Item*

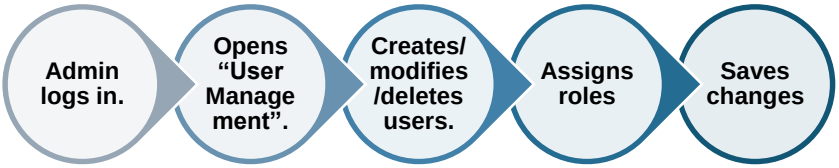
- **Actors:** Storekeeper



- **Outcome:** Accurate stock update and issuance record.

Scenario 4: *Managing Users*

- **Actors:** Admin



- **Outcome:** Only authorized users access the system.

3.7 Use Case Model

The Use Case Model describes the interaction between users (actors) and the system. It ensures the system captures all required functionalities.

3.7.1 Actor identification

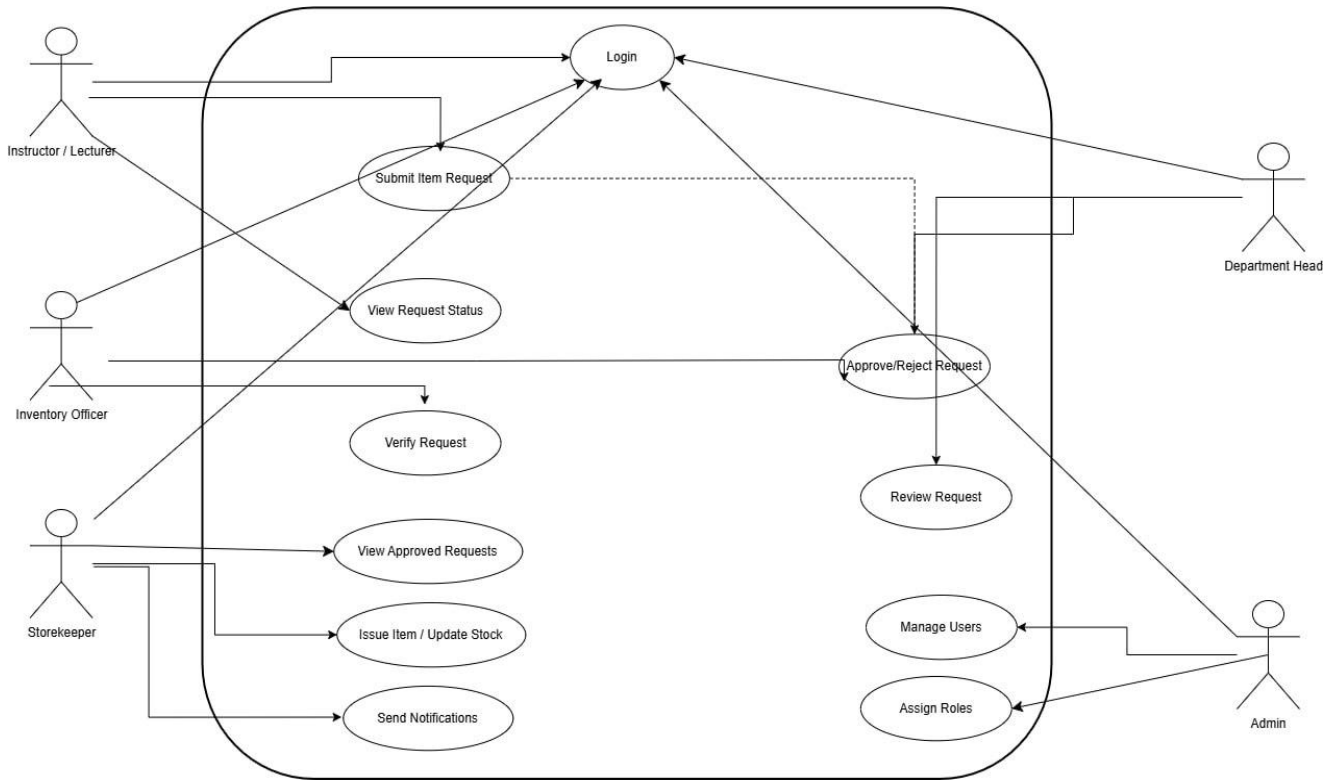
Actor	Description
Instructor / lecturer	Initiates inventory item requests.
Department head	Reviews and approves/rejects requests.
Inventory officer	Verifies approved requests and ensures accuracy.
Storekeeper (main store)	Issues non-consumable items and updates stock.
Department storekeeper	Issues consumable items within departments.
Admin	Manages system settings, roles, users, backups.

3.7.2 Use Case Identification

- ☑ UC-01: Login to System
- ☑ UC-02: Request Inventory Item
- ☑ UC-03: Approve/Reject Request (Department Head)
- ☑ UC-04: Verify Request Accuracy (Inventory Officer)
- ☑ UC-05: Issue Item to Requester (Storekeeper)
- ☑ UC-06: Update Stock Levels
- ☑ UC-07: Manage Users & Roles (Admin)

- ☑ UC-08: Manage Inventory Items
- ☑ UC-09: Generate Reports
- ☑ UC-10: Receive Notifications
- ☑ UC-11: Track Item History
- ☑ UC-12: Manage Suppliers (optional)

3.7.3 Use Case Diagram



3.7.4 Use Case Descriptions

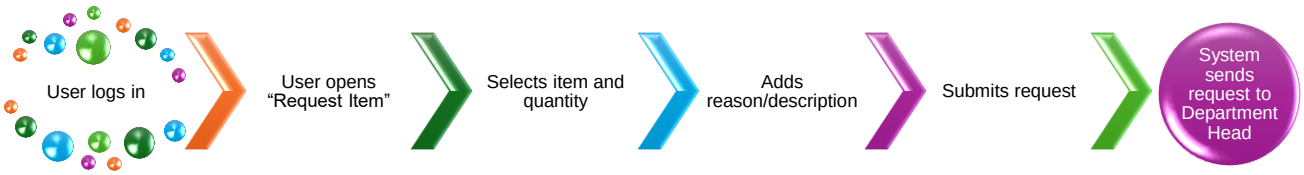
<u>Use Case 1</u>: Submit Item Request

- **Primary Actor**: Instructor / Lecturer
- **Supporting Actors**: Department Head, System
- **Description**: The user submits a digital request for an item needed by the department.
- **Preconditions**: User is logged in, Item exists in the system
- **Post conditions**: Request status saved as "Pending department head Approval"

- **Alternate Flow**: If item does not exist then, System shows error.

  If quantity exceeds available stock then, System shows warning.

Main Flow:



<u>Use Case 2</u>: Approve/Reject Request (Department Head)

- **Primary Actor**: Department Head
- **Supporting Actors:** Instructor, Inventory Officer, System
- **Description**: Department Head reviews the submitted request and approves or rejects it.
- **Preconditions**: Request exists, User has Department head role
- **Post conditions**: Status updated (Approved by DH / Rejected)
- **Alternate Flow**: If user tries to approve without permission then, System denies.

Main Flow:

<u>Use Case 3:</u> Verify Request (Inventory Officer)

- **Primary Actor**: Inventory Officer
- **Supporting Actors**: Department Head, Storekeeper
- **Description**: Inventory Officer verifies availability and correctness of approved requests.
- **Preconditions**: Request approved by Department Head
- **Post conditions**: Status = "Approved for Issuance"
- **Alternate Flow**: If item is out of stock → Reject request.

Main Flow:

<u>Use Case 4:</u> Issue Inventory Item

- Primary Actor: Storekeeper
- Supporting Actors: Inventory Officer, Instructor
- Description: Storekeeper issues items after final approval.
- Preconditions: Request approved, Stock is available
- Post conditions: Stock updated, Issuance record saved
- Alternate Flow: User attempts to return unrelated item then, System rejects

Main Flow:

<u>Use Case 5:</u> Manage Inventory Items

- Primary Actor: Admin / Store Manager
- Description: Admin adds, edits, or removes inventory items.
- Preconditions: User has Admin privileges
- Post conditions: Item record updated in database
- Alternate Flow: Invalid data → system shows error.

Main Flow:

<u>Use Case 6:</u> Update Stock Levels

- **Primary Actor**: Storekeeper / Inventory Officer
- **Description**: Stock is manually adjusted after audits, returns, or corrections.
- **Preconditions**: User has permission
- **Post conditions**: Stock level updated and logged

Main Flow:

<u>Use Case 7:</u> Manage Users & Roles

- **Primary Actor**: Admin
- **Description**: Admin creates, updates, or removes system users.
- **Preconditions**: Admin must be authenticated
- **Post conditions**: User account created or modified

Main Flow:

<u>Use Case 8:</u> Generate Inventory Reports

- **Primary Actor**: Inventory Officer / Admin
- **Description**: Generate summary, monthly, or item movement reports.
- **Preconditions**: System has stored records
- **Post conditions**: Report downloadable
- **Alternate Flow**: No data → System shows empty report.

Main Flow:

Use Case 9: Track Item Return

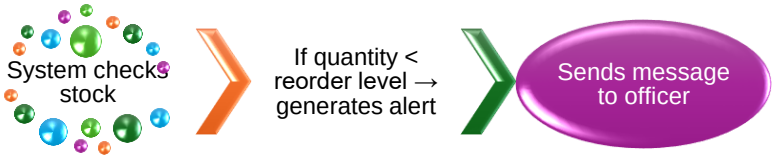
- Primary Actor: Instructor / Storekeeper
- Description: Items (especially non-consumables) are returned to store.
- Preconditions: Item previously issued
- Post conditions: Return recorded

Main Flow:

Use Case 10: Low Stock Notification

- Primary Actor: System
- Supporting Actors: Inventory Officer, Storekeeper
- Description: System automatically notifies officers when quantity drops below threshold.
- Preconditions: Reorder level defined
- Post conditions: Alert stored, notification delivered

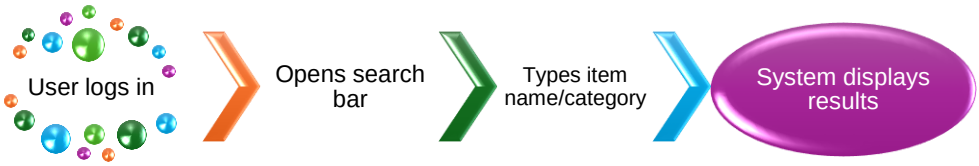
Main Flow:



Use Case 11: Search Inventory Items

- Primary Actor: Instructor / Storekeeper / Department Head
- Description: User searches the inventory list.
- Preconditions: Items exist
- Post conditions: Search results displayed

Main Flow:



<u>Use Case 12</u>: Backup and Restore System Data

- Primary Actor: Admin
- Description: Admin performs system backup or restores previous data.
- Preconditions: Admin authenticated
- Post conditions: Backup created OR previous data restored

Main Flow:

3.8 Object Model

The Object Model represents the **static structure** of the Inventory Management System (IMS). It defines the important data entities (objects), their attributes, relationships, and the structure of the database. This model ensures consistency between analysis, design, and implementation, and acts as a foundation for building the system's database schema and class structure.

3.8.1 Data Dictionary

It provides detailed definitions of all major data entities, their attributes, descriptions, types, constraints, and relationships.

A. USER TABLE (instructor/lecturer /any user)

Attribute	Type	Description	Constraints
user_id	INT	Unique ID for each user	Primary Key, Auto Increment
full_name	VARCHAR(100)	User's full name	Not Null
email	VARCHAR(120)	Login email	Unique, Not Null
password	VARCHAR(255)	Encrypted password	Not Null
role	ENUM ('Admin', 'Storekeeper', 'Inventory Officer', 'Department Head', 'Instructor')	Defines user permissions	Not Null
department	VARCHAR(100)	User's department	Optional

B. INVENTORY ITEM TABLE

Attribute	Type	Description	Constraints
item_id	INT	Unique ID for each item	Primary Key
item_name	VARCHAR(150)	Name of the item	Not Null
category	ENUM(“Consumable”,“Non-Consumable”)	Item category	Not Null
description	TEXT	Details about the item	Optional
quantity	INT	Current stock quantity	Not Null
unit	VARCHAR(50)	Unit of measurement	e.g., pcs, box
location	VARCHAR(100)	Store or lab location	Not Null
reorder_level	INT	Minimum quantity before alert	Not Null
supplier_id	INT	Linked supplier	Foreign Key

C. REQUEST TABLE

Attribute	Type	Description	Constraints
request_id	INT	Unique ID for each request	Primary Key
requester_id	INT	User making the request	Foreign Key (User)
item_id	INT	Requested item	Foreign Key
quantity	INT	Amount requested	Not Null
reason	TEXT	Purpose of request	Not Null
status	ENUM('Pending', 'Approved by Dept Head',' Approved by Inventory Officer', 'Rejected', 'Issued')	Approval status	Default: Pending
created_at	DATETIME	Request submission	Default: CURRENT_TIMESTAMP

D. APPROVAL TABLE

Attribute	Type	Description	Constraints
approval_id	INT	Unique ID	Primary Key
request_id	INT	Request being approved	Foreign Key
approved_by	INT	Approver user ID	Foreign Key
approval_role	ENUM('Department Head','Inventory Officer')	Approval stage	Not Null
decision	ENUM('Approved','Rejected')	Approval decision	Not Null
timestamp	DATETIME	Date of approval	Default: CURRENT_TIMESTAMP

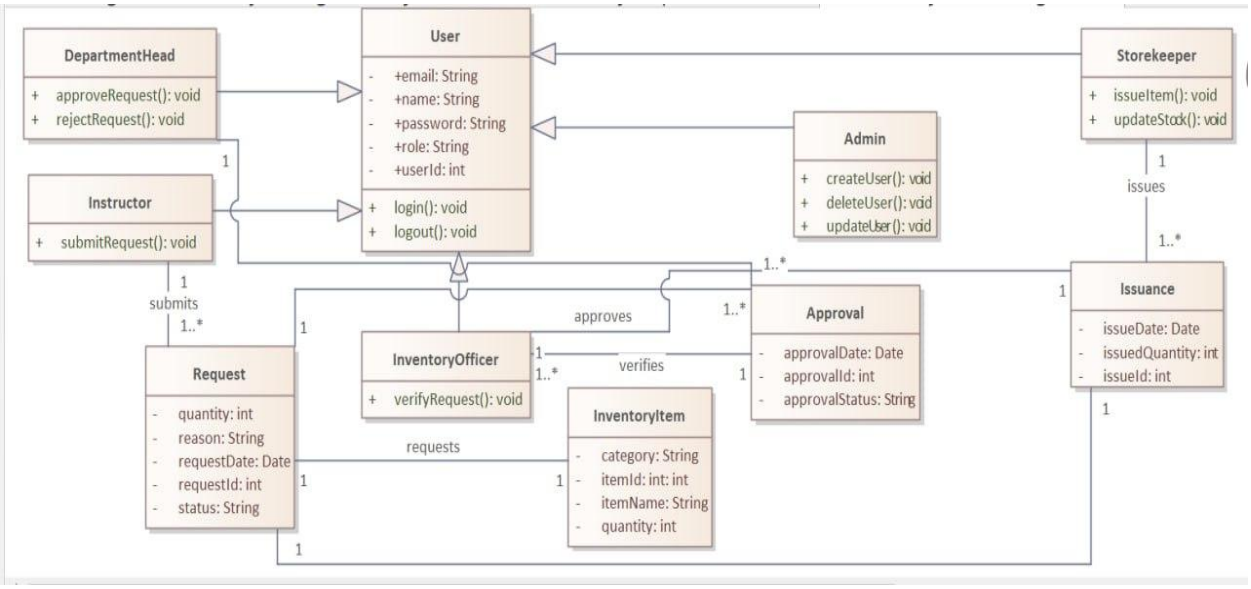
E. ISSUANCE TABLE

Attribute	Type	Description	Constraints
issuance_id	INT	Unique issuance operation	Primary Key
request_id	INT	Related approved request	Foreign Key
storekeeper_id	INT	User issuing the item	Foreign Key
issued_quantity	INT	Quantity issued	<= requested
issue_date	DATETIME	Date issued	Default: CURRENT_TIMESTAMP

F. NOTIFICATION TABLE

Attribute	Type	Description	Constraints
notification_id	INT	Unique notification	Primary Key
user_id	INT	Receiver	Foreign Key
message	TEXT	Notification content	Not Null
status	ENUM('Read','Unread')	Read status	Default: Unread
created_at	DATETIME	Time sent	Default: CURRENT_TIMESTAMP

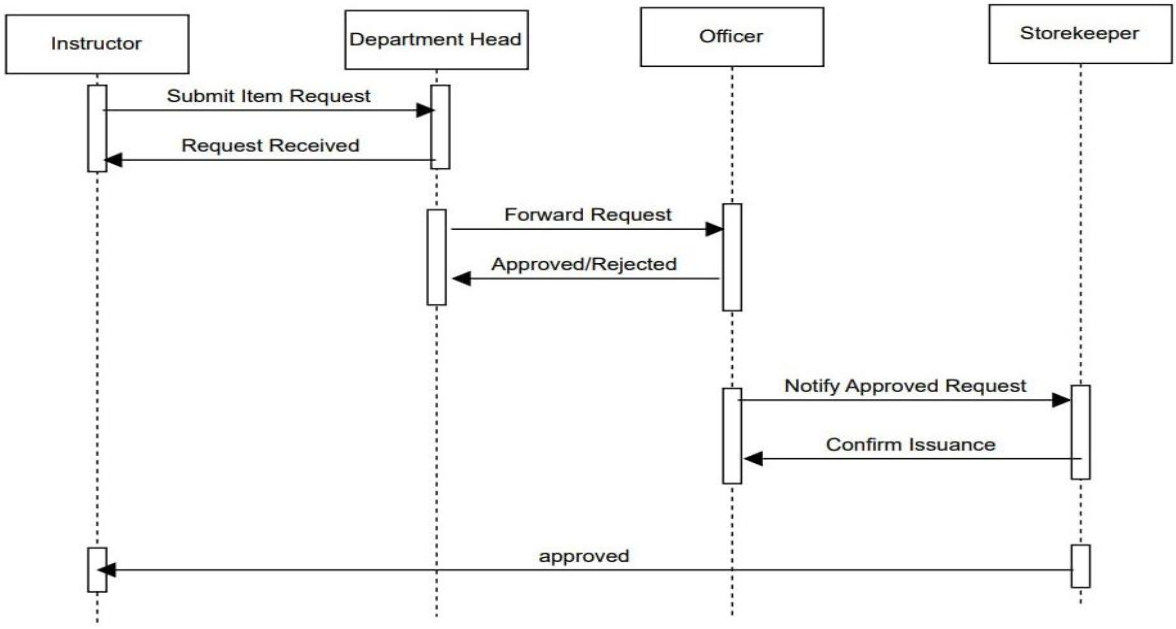
3.8.2 Class Diagram



3.9 Dynamic Models

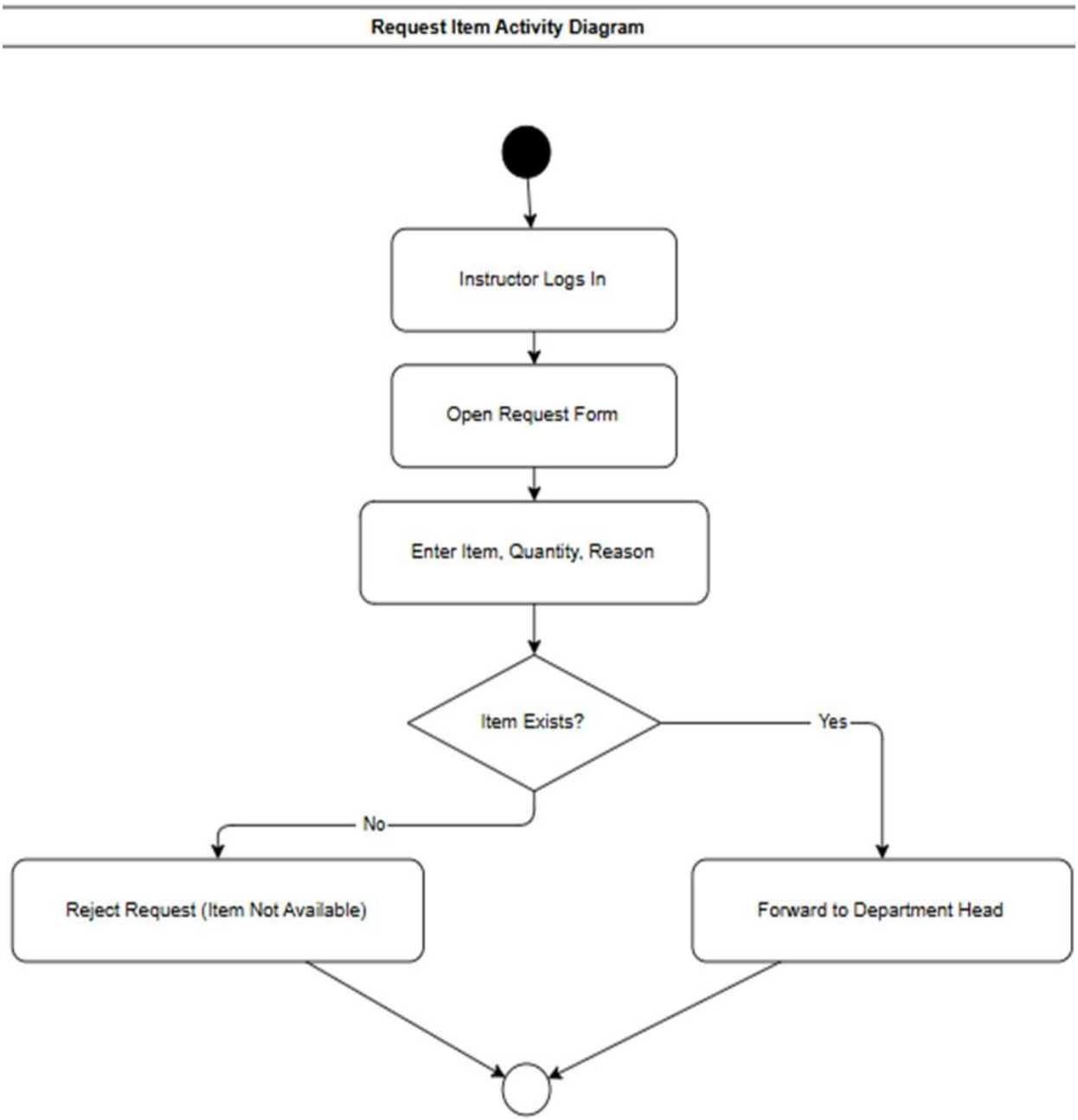
Dynamic models describe **how the proposed Inventory Management System behaves over time**, especially how different components and actors interact. The dynamic behavior focuses on **request workflow, approval workflow, item issuance, and stock update processes**.

3.9.1 Sequence Diagrams

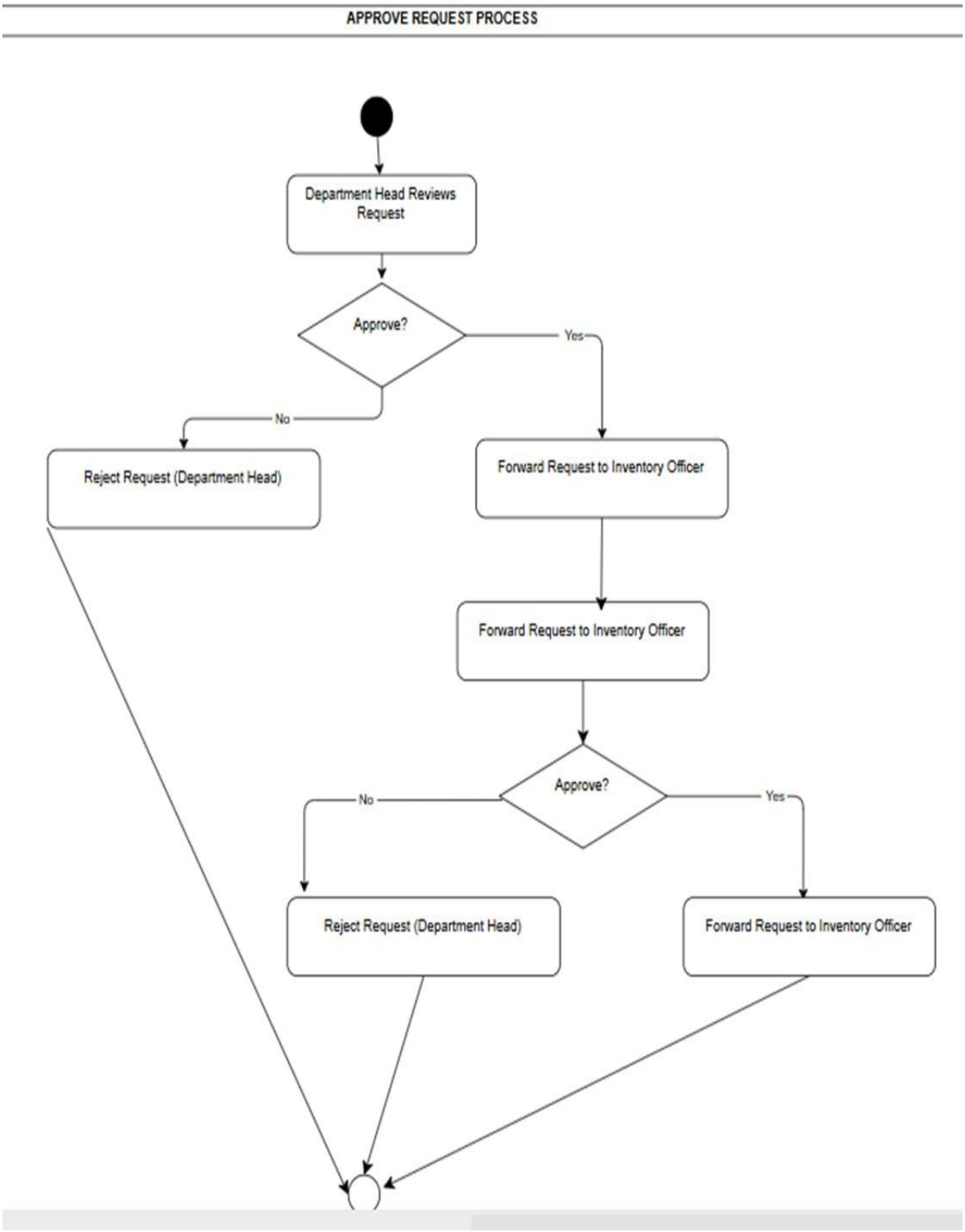


3.9.2 Activity Diagrams

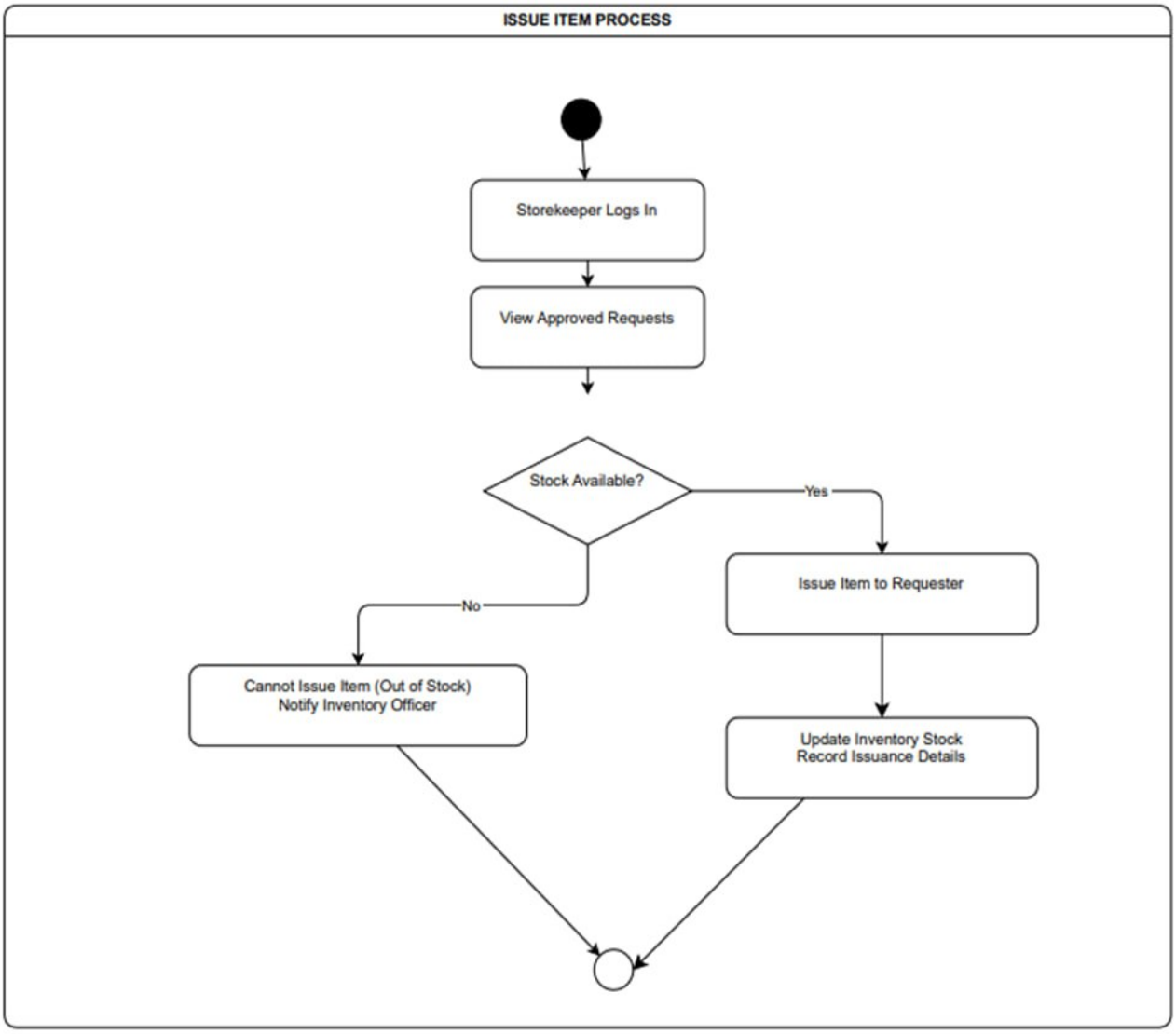
Request item activity diagram



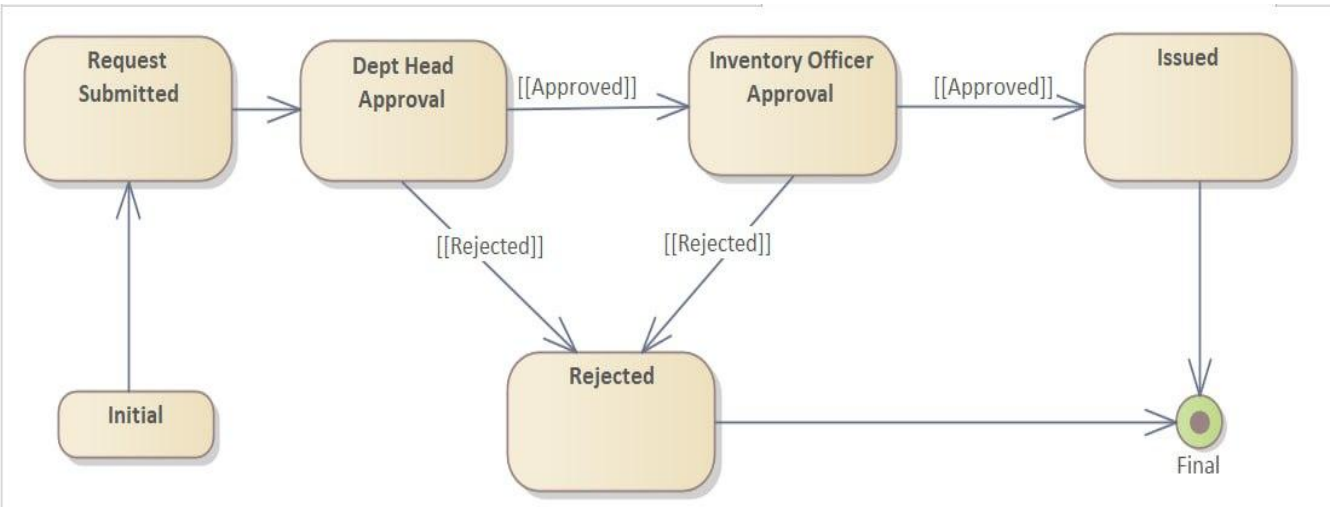
Approval request activity diagram



Issue item activity diagram



3.9.3 State Chart Diagrams



CHAPTER FOUR: SYSTEM DESIGN

4.1 Overview

The System Design chapter provides the architectural blueprint for the proposed **Inventory Management System (IMS)** of Adama Science and Technology University (ASTU). The design outlines how the system's components will be structured, how they will interact, and how user requirements will be translated into a technically workable system.

Since the current university workflow is fully manual and the previous digital system failed due to poor usability, this design focuses on creating a **simple, user-friendly, scalable, and reliable system** that automates all inventory processes including item requests, approvals, issuance, stock monitoring, and reporting.

The design covers the major structural views of the system, including:

- System architecture
- Subsystem decomposition
- Data management structure
- Component and deployment models
- Access control design
- User interface structure

4.2 Purpose of System Design

The purpose of the System Design is to transform the gathered functional and non-functional requirements into a clear and testable technical structure. Specifically, the design aims to:

- ✓ Guide Developers
- ✓ Ensure System Quality
- ✓ Clarify System Components
- ✓ Reduce Implementation Errors

4.3 Design Goals

The Inventory Management System is designed to meet the following goals:

1. Usability

The previous system was abandoned because it was difficult to use. The new system must be **simple, intuitive, and user-friendly** for storekeepers, lecturers, department heads, and inventory officers.

2. Accuracy and Data Integrity

Ensure that all stock transactions (request, approval, issuance, return) are: recorded correctly, timestamped, traceable, consistent across all stores.

3. Automation of the Workflow

Replace all manual paper-based procedures with: digital request submission, automated approval chain, automatic notifications, real-time status tracking.

4. Real-Time Stock Visibility

Both Main Store and Department Stores must share synchronized, live inventory updates to prevent: stock-outs, duplication, inaccurate reports.

5. Security and Access Control

Provide strict role-based restrictions: Each role sees only what is relevant to them.

- Admin → manages system
- Inventory Officer → verifies items
- Department Head → approves
- Storekeeper → issues items
- Instructor → requests items

6. Reliability and Fault Tolerance

The system must be accessible during university working hours and resilient to data loss through:

- ✓ automatic backups
- ✓ stable database design
- ✓ recovery mechanisms

7. Maintainability

Designed with modularity so that new features can be added easily without affecting existing modules.

8. Performance

- ✓ Page response time < 3 seconds
- ✓ Handle multiple concurrent users
- ✓ Efficient queries for large inventory records

9. Scalability

Structure the system to support:

- ✓ growing number of items
- ✓ additional inventory categories
- ✓ future integration with ERP or procurement systems

10. Compliance and Auditability

System logs must support:

- ✓ accountability
- ✓ auditing
- ✓ tracking item history
- ✓ identifying responsible persons

4.4 Proposed System Architecture

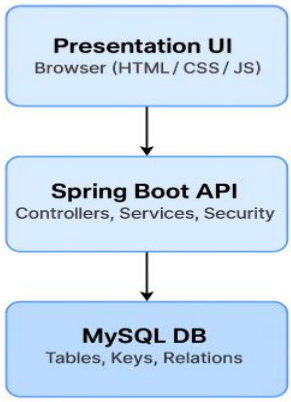
The IMS will follow a **Three-Tier Architecture**, which separates the application into three logical layers:

1. Presentation Layer (Client/UI Layer)

- Implemented using HTML, CSS, and basic JavaScript.
- Contains login screen, request forms, approval screens, stock dashboard, admin panels.
- Communicates with backend using HTTP/REST.

2. Application Layer (Business Logic Layer)

- Developed using **Java Servlet**.
- Handles authentication, request workflows, approval logic, issuance logic, reports generation. Enforces business rules collected from stakeholders.



3. Data Layer (Database Layer)

- Uses **MySQL**.
- Stores users, items, inventory logs, requests, approvals, and audit trails.
- Ensures data consistency across stores and departments.

4.4.1 System Processes

The system includes the following major processes:

1. **User Authentication Process**
    - Login → Role verification → Access granted.
2. **Inventory Request Process**
    - Instructor submits request → DH approves → Inventory Officer verifies → Storekeeper issues → Stock updated.

3. **Stock Management Process**
    - Add new item
    - Modify item details
    - Update quantities
    - Track issued/returned/damaged items.
4. **Reporting Process**
    - Generate weekly/monthly reports
    - Stock valuation
    - Request history
5. **Notification Process**
    - Alerts for low stock
    - Alerts for new approvals
    - Issuance confirmation

4.4.2 Subsystem Decomposition and Description

The system is divided into the following subsystems:

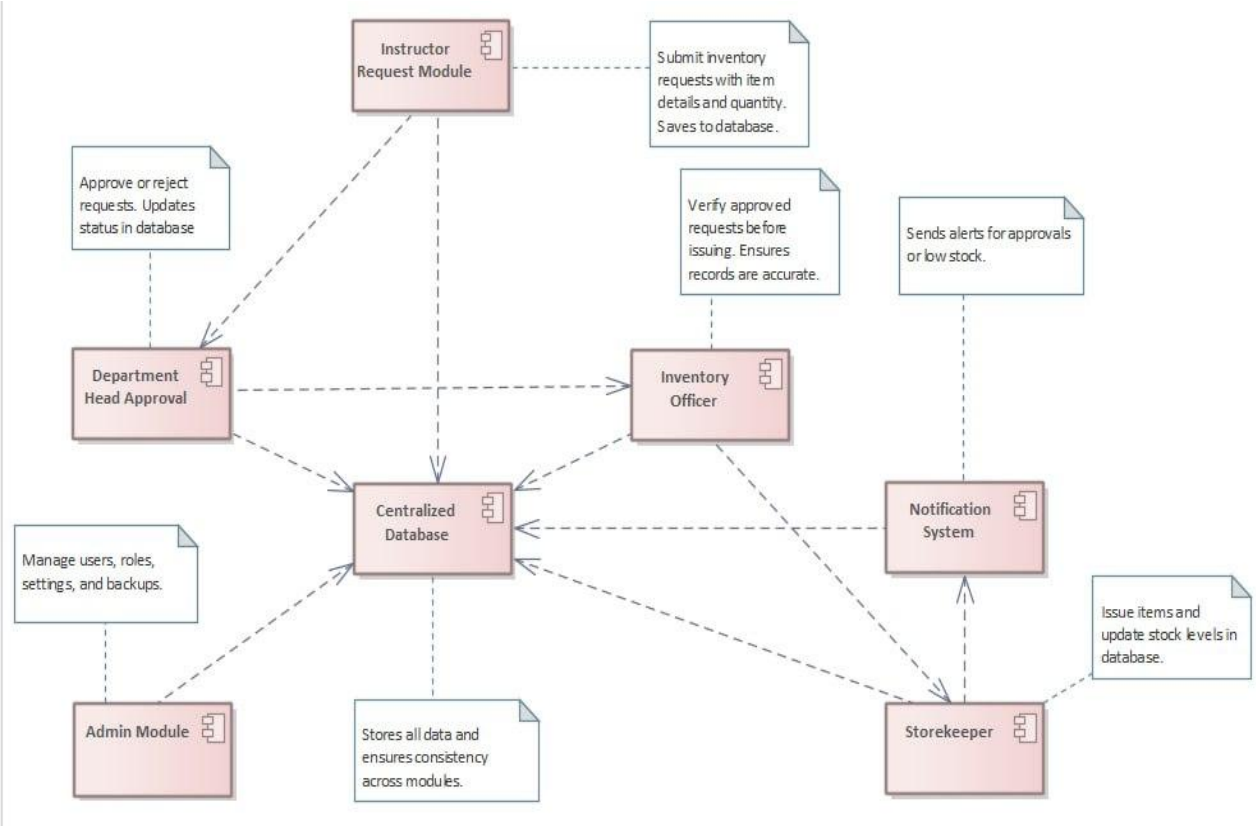
Subsystem	**Description**
Authentication Subsystem	Manages login, permissions, and roles.
Inventory Subsystem	Manages items, categories, quantities, availability.
Request & Approval Subsystem	Handles workflows for requesting, approving, and issuing items.
User Management Subsystem	Admin adds users, roles, departments.
Reporting Subsystem	Generates reports for issued items, low stock, monthly summaries.
Notification Subsystem	Sends alerts and status updates.

4.4.3 Hardware–Software Mapping

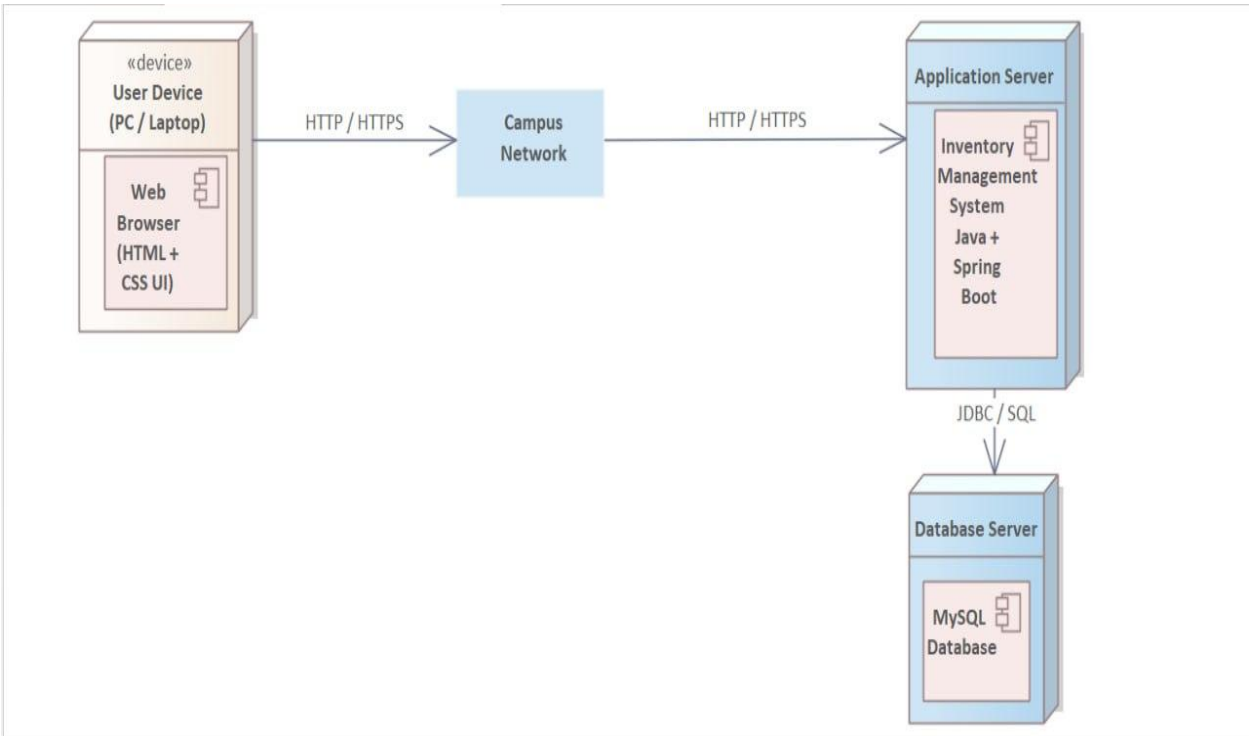
Hardware Requirements: - Server machine or cloud server, Workstation devices (instructors, storekeepers), LAN or WiFi connections.

Software Requirements: - Java JDK, Servlet Framework, MySQL Database, Web Browser (Chrome recommended), Git/GitHub for version control.

4.4.4 Component Diagram



4.4.5 Deployment Diagram



4.4.6 Database Design (ERD, Tables, Keys)

The IMS uses a **relational database** to store, manage, and retrieve inventory, user, and transaction data in a structured and consistent manner.

The database design ensures:

- Data integrity and accuracy
- Reduced data redundancy
- Efficient querying and reporting
- Secure access control

The database supports both **consumable** and **non-consumable** items and manages workflows such as **item requests, approvals, issuance, and stock updates**.

Key Relationships

- ✓ A **User** belongs to one **Department**
- ✓ A **User** has one **Role**
- ✓ A **Department** submits many **Requests**
- ✓ A **Request** contains one or more **Items**
- ✓ A **Request** goes through an **Approval**
- ✓ Approved requests result in **Item Issuance**
- ✓ **Inventory** tracks quantities of **Items**
- ✓ **Notifications** are sent to users during workflow changes

4.4.7 Access Control & Security Design

Security features include:

- **Role-based Access Control (RBAC)**
- **Authentication**
  - Username + password
  - Secure password hashing
- **Authorization**
  - Only allowed roles access specific modules.

- **Data Security**
    - HTTPS
    - Input validation
    - Audit logs
    - Restricted DB access

4.5 Risk Assessment & Mitigation

Risk	Impact	Likelihood	Mitigation
User resistance to new system	High	Medium	Provide training, simple UI
Server failure	High	Low	Regular backups, cloud hosting
Data loss	High	Low	Automatic DB backups
Incorrect stock values	Medium	Medium	Strict validation, audit trails
Security breach	High	Low	Strong authentication, encryption

4.6 System Integration Plan

We will build a web application by using:

- ✓ Frontend: HTML, CSS to Provides a structured, interactive, and visually appealing user interface.
- ✓ Backend: Java + Servlet Implements business logic and manages server-side operations efficiently.
- ✓ Database: SQL – Stores structured data and ensures secure retrieval of stocks.

The integration plan defines how all modules will be combined:

Phase 1 – Backend Setup

- Create database schema

Phase 2 – Authentication & Roles

- Implement login functionality

Phase 3 – Inventory Module Integration

- Add items, Update stock
- Connect with inventory officer's dashboard

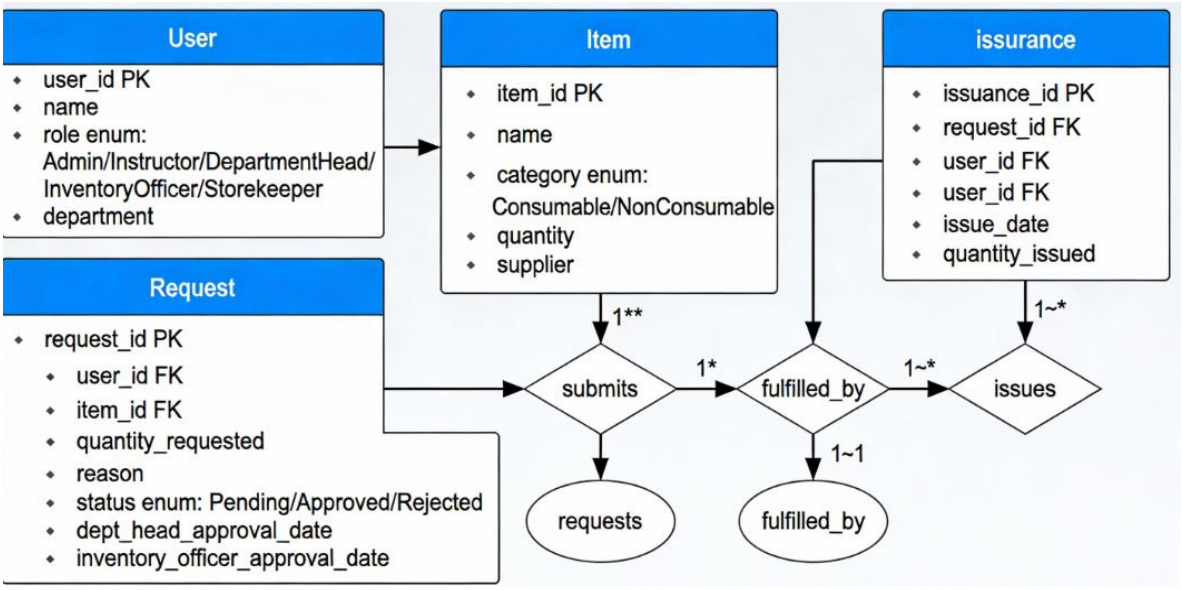
Phase 4 – Request & Approval Workflow Integration

- Instructor → Dept Head → Inventory Officer → Storekeeper
- Test full workflow

Phase 5 – Reporting & Notification Integration

- Weekly/monthly reports
- Email/SMS notifications

Phase 6 – System Testing & Deployment



CHAPTER FIVE: TESTING & VALIDATION

This chapter outlines the testing strategy, methods, and specific test cases intended to verify that the Inventory Management System meets the functional and non-functional requirements defined in previous chapters. The goal is to ensure the system is reliable, secure, and ready for deployment within the university network.

5.1 Testing strategy

The testing strategy for the IMS follows a **planned and systematic approach**, where testing activities are performed throughout the system development lifecycle. Both **manual testing** and **scenario-based testing** are applied to verify functional and non-functional requirements.

The strategy includes:

- Testing individual system components.
- Testing interactions between modules.
- Validating system behavior against the SRS requirements.
- Involving end users to confirm system usability and acceptance.

Testing is conducted using predefined test cases, realistic test data, and documented expected results.

5.2 Test Case Design

Test Case 1: User Authentication (Login)

- ❖ Objective: Verify role-based access control.
- ❖ Precondition: User is registered in the database.

Step	Action	Expected Result
1	Enter valid Admin ID & Password.	Redirect to Admin Dashboard.
2	Enter valid Instructor ID & Password.	Redirect to Request Form/instructor dashboard
3	Enter valid Storekeeper ID & Password.	Redirect to Stock Dashboard.
4	Enter invalid credentials.	Display "Invalid Username or Password" error.

Test Case 2: Submit Inventory Request (Instructor)

- Objective: Verify the "Request Initiation" workflow.
- Precondition: Instructor is logged in; Item exists in DB.

Step	Action	Expected Result
1	Select "Dell Monitor" from dropdown.	System displays current stock availability.
2	Enter Quantity and Reason.	Button becomes active.
3	Click "Submit."	System displays "Request Sent to DH" success message.
4	Check Database.	New record created with status PENDING.

Test Case 3: Request Approval (DH & Inventory Officer)

- Objective: Verify the multi-step approval workflow.
- Precondition: A pending request exists.

Step	Action	Expected Result
1	DH logs in & views requests.	Pending request from Instructor is visible.
2	DH clicks "Approve."	Status changes to APPROVED_BY_DH.
3	Inventory Officer logs in.	Request is visible for verification.
4	Inventory Officer verifies & clicks "Approve."	Status changes to APPROVED. Notification sent to Storekeeper.

Test Case 4: Issue Item (Storekeeper)

- Objective: Verify stock deduction and issuance.
- Precondition: Request status is APPROVED.

Step	Action	Expected Result
1	Storekeeper clicks "Issue Item."	System changes status to ISSUED.
2	Check Inventory Database.	Stock count for "Dell Monitor" decreases by 1.
3	Check Report.	Transaction recorded in "Issued Items" log.

5.3 Test Data

Test data mirrors real inventory items: non-consumable (computers, lab equipment) and consumable (markers, papers, toners) with sample quantities, categories, and suppliers. User data includes test accounts for all roles with predefined permissions; request data features varying quantities, reasons, and statuses (Pending, Approved, Rejected).

5.4 System Testing

System testing evaluates the application as a complete entity.

Performance Testing:

- ✓ Requirement: the system must respond within 3 seconds.
- ✓ Test: Load the dashboard with 50 concurrent users (simulating a busy day at ASTU) and measure page load time.

Security Testing:

- ✓ Requirement: Secure Login.
- ✓ Test: Attempt to access the /admin/settings URL while logged in as an Instructor. The system must deny access (403 Forbidden).

Compatibility Testing:

- ✓ Test: Open the web application on Chrome, Firefox, and Edge to ensure the HTML/CSS renders correctly on all browsers.

5.5 User Acceptance Testing (UAT)

User Acceptance Testing is conducted to ensure that the system meets the expectations and needs of actual users.
Key stakeholders such as **storekeepers, department heads, inventory officers, and instructors** participate in this phase.

- Users perform real-life tasks using the system.
- Feedback is collected regarding usability, clarity, and workflow.
- The system is evaluated for ease of use and practical applicability.

CHAPTER SIX: IMPLEMENTATION & DEPLOYMENT PLAN

6.1 Implementation Approach

The IMS for ASTU will be developed using a modular, iterative approach. The system will be built in phases aligned with the Software Development Life Cycle (SDLC):

Requirement Analysis: Completed through stakeholder interviews and workflow documentation.

Design: Includes UI mockups, ER diagrams, and use case modeling.

Development: Frontend using HTML/CSS, backend using Java with Servlet, and MySQL for database management.

Testing: Unit testing, integration testing, and user acceptance testing will be conducted.

Deployment: Final system will be deployed on the university's internal network.
Version control will be managed via Git/GitHub to ensure collaborative development and rollback capability.

6.2 Deployment Environment

The IMS will be deployed in a secure, campus-wide environment with the following specifications:

- ✓ **Server OS**: Ubuntu Linux or Windows Server
- ✓ **Application Server**: Servlet (Servlet embedded)
- ✓ **Database Server**: MySQL
- ✓ **Client Access**: Modern browsers (Chrome, Firefox, Edge)
- ✓ **Network**: ASTU internal LAN with role-based access control
- ✓ **Security**: HTTPS, firewall protection, and user authentication via password hashing

6.3 Installation Procedure

The installation will follow these steps:

1. **Server Setup**: Install Java JDK, MySQL, and configure Servlet runtime.
2. **Database Initialization**: Import schema and seed data using SQL scripts.
3. **Application Deployment**: Deploy Servlet application via WAR/JAR file.
4. **Configuration**: Set environment variables, database credentials, and system parameters.
5. **Testing**: Run smoke tests to verify installation integrity.
6. **Access Provisioning**: Create user accounts and assign roles (Admin, Storekeeper, etc.)

6.4 Training Plan for Users

To ensure smooth adoption, a structured training plan will be implemented:

Target Users: Storekeepers, Inventory Officers, Department Heads, Instructors, Admins

Training Format: Hands-on workshops, user manuals, and video tutorials

Topics Covered:

- ✓ Logging in and navigating the UI
- ✓ Submitting and approving requests
- ✓ Issuing items and updating stock
- ✓ Generating reports and managing users

6.5 Maintenance & Support Plan

To ensure long-term reliability and usability, the following maintenance and support strategies will be adopted:

- **Routine Maintenance**: Weekly database backups, monthly performance checks, and quarterly updates
- **Bug Fixes**: Logged via GitHub Issues and resolved in scheduled sprints
- **User Support**: Dedicated support team reachable via email or internal ticketing system
- **System Monitoring**: Real-time logs and alerts for system errors or downtime
- **Scalability**: Modular design allows future expansion (e.g., mobile access)

CHAPTER SEVEN: CONCLUSION AND RECOMMENDATIONS

7.1 Summary of Work

This project focused on analyzing, designing, and documenting the requirements for a digital Inventory Management System (IMS) for Adama Science and Technology University (ASTU). Through stakeholder interviews, workflow observations, and analysis of the existing manual process, the team identified major challenges such as slow request processing, lack of real-time stock visibility, inconsistent record-keeping, and poor usability of the previously developed system.

The team prepared a comprehensive SRS document that includes functional and non-functional requirements, use case descriptions, workflow diagrams, and database design. The proposed system introduces a centralized digital platform that supports item requests, approvals, stock management, reporting, and role-based access. The project also outlined the implementation plan, testing strategy, and deployment approach to ensure the system can be developed and adopted effectively.

Overall, the work completed lays a strong foundation for building a modern, efficient, and user-friendly inventory system that addresses the university's operational needs.

7.2 Limitations of the Project

Despite the progress made, the project has several limitations:

- **Limited Stakeholder Interviews**: Only a few key personnel (storekeeper, lab assistant) were interviewed. Additional insights from department heads, procurement officers, and administrators would strengthen requirement accuracy.
- **Incomplete Real-World Data**: Some inventory processes vary across departments, and full documentation of all variations was not possible within the project timeline.

- **Prototype Not Fully Developed**: While requirements and design are well-defined, the actual system implementation is still in progress.
- **No Integration Testing with Existing Systems**: The University may have other digital systems (procurement, finance) that were not evaluated for integration.
- **Usability Testing Pending**: Since the UI is not fully implemented, real user feedback on interface design is not yet available.

These limitations highlight areas that require further work during the development and deployment phases.

7.3 Future Improvements

To enhance the system and ensure long-term sustainability, the following improvements are recommended:

- **Full Automation of Procurement Workflow**
  Integrate purchase requests, supplier management, and budget approvals into the system.
- **Mobile Application Support**
  Provide mobile access for instructors and storekeepers to submit or approve requests on the go.
- **Barcode or QR Code Integration**
  Automate item identification, stock counting, and movement tracking.
- **Advanced Reporting & Analytics**
  Add dashboards for consumption trends, stock forecasting, and audit summaries.
- **Integration with University Systems**
  Connect the IMS with finance, HR, and procurement systems for seamless data flow.
- **Cloud Deployment**
  Consider migrating to a cloud environment for scalability, reliability, and remote access.
- **Enhanced Security Features**
  Implement two-factor authentication, audit logs, and role-based permission refinement.

These improvements will help the system evolve into a more robust and intelligent inventory platform.

7.4 Final Recommendations

Based on the findings and the proposed solution, the following recommendations are made:

- **Adopt the Digital System Gradually**
  Begin with pilot departments to ensure smooth transition from manual to digital workflows.
- **Provide Continuous Training**
  Regular training sessions should be conducted to ensure all users understand the system and use it effectively.
- **Assign a Dedicated System Administrator**
  A trained staff member should manage user accounts, backups, and system configurations.
- **Ensure Regular Maintenance and Updates**
  The system should be monitored, updated, and improved continuously to maintain performance and security.
- **Encourage User Feedback**
  Collect feedback from instructors, storekeepers, and department heads to refine the system over time.
- **Promote a Culture of Digital Adoption**
  University leadership should encourage staff to use the system consistently to eliminate reliance on paper-based processes.

References

The following documents, standards, and technical resources were referenced in the preparation of this Software Requirements Specification (SRS):

1. Servlet Documentation. Available at:
https://docs.spring.io/springboot/docs/current/reference/html/ (Used for Backend logic).

2. MySQL 8.0 Reference Manual. Oracle Corporation. Available at:
https://dev.mysql.com/doc/refman/8.0/en/ (Used for Database Design).

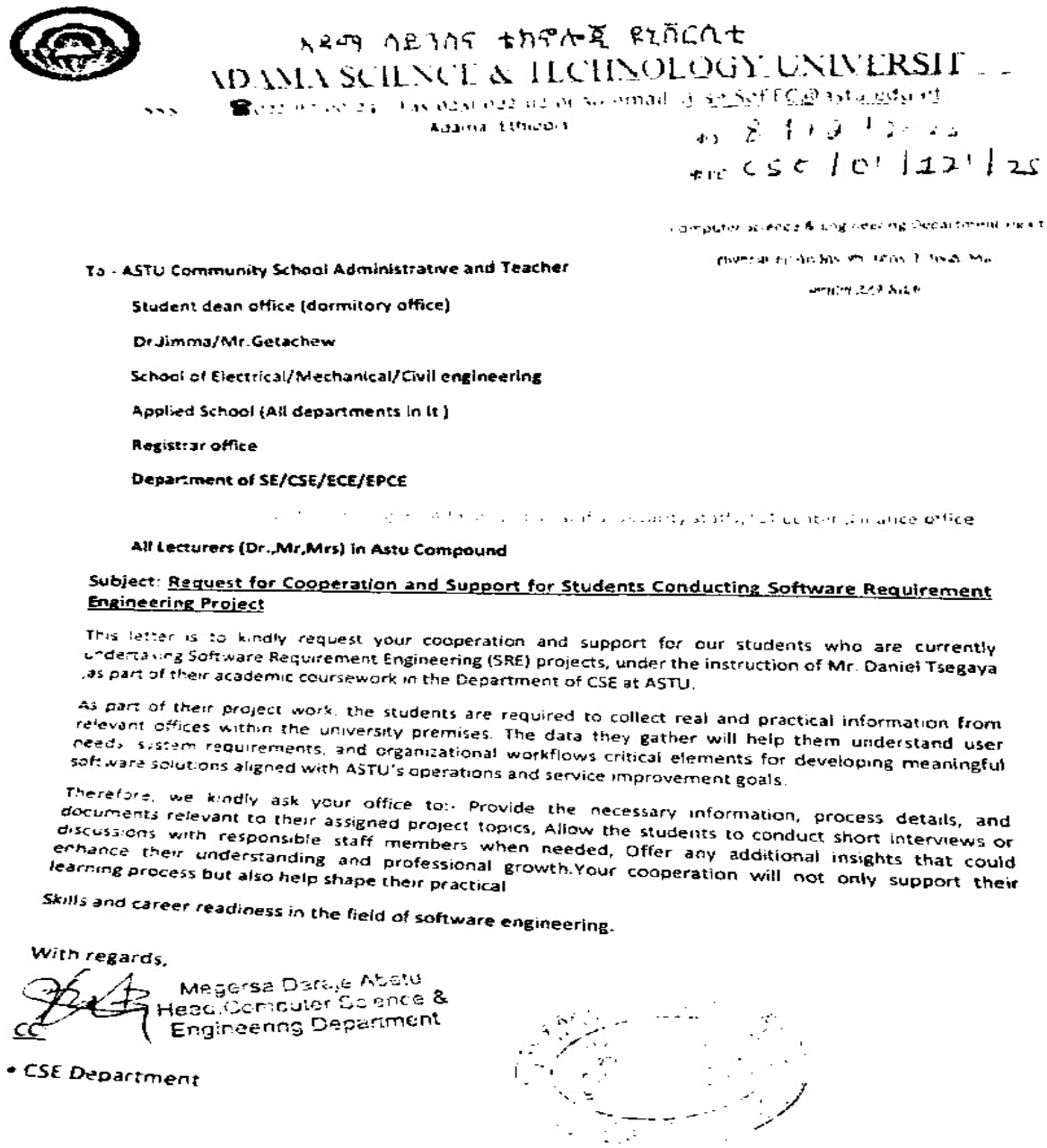
3. MDN Web Docs (Mozilla). HTML & CSS Developer Guide. Available at:
https://developer.mozilla.org/ (Used for Frontend Interface).

4. ASTU Group Project Progress Report (Inventory Management System). Information Collected from Stakeholders (Lab Assistant and Main Storekeeper). Unpublished Group Report, 2024.

5. Adama Science and Technology University. Course Outline: Software Requirements Engineering, School of Electrical Engineering & Computing.

Appendices

APPENDIX A: INTERVIEW/OBSERVATION DATA



አዳማ ሳይንስና ቴክኖሎጂ ዩኒቨርሲቲ

ADAMA SCIENCE & TECHNOLOGY UNIVERSITY

Adama, Ethiopia

ቁጥር CSE/01/121/25

To - ASTU Community School Administrative and Teacher

Student dean office (dormitory office)

Dr.Jimma/Mr.Getachew

School of Electrical/Mechanical/Civil engineering

Applied School (All departments in it)

Registrar office

Department of SE/CSE/ECE/EPCE

All Lecturers (Dr.,Mr,Mrs) in Astu Compound

Subject: Request for Cooperation and Support for Students Conducting Software Requirement Engineering Project

This letter is to kindly request your cooperation and support for our students who are currently undertaking Software Requirement Engineering (SRE) projects, under the instruction of Mr. Daniel Tsegaya ,as part of their academic coursework in the Department of CSE at ASTU.

As part of their project work, the students are required to collect real and practical information from relevant offices within the university premises. The data they gather will help them understand user needs, system requirements, and organizational workflows critical elements for developing meaningful software solutions aligned with ASTU's operations and service improvement goals.

Therefore, we kindly ask your office to:- Provide the necessary information, process details, and documents relevant to their assigned project topics, Allow the students to conduct short interviews or discussions with responsible staff members when needed, Offer any additional insights that could enhance their understanding and professional growth.Your cooperation will not only support their learning process but also help shape their practical

Skills and career readiness in the field of software engineering.

With regards,

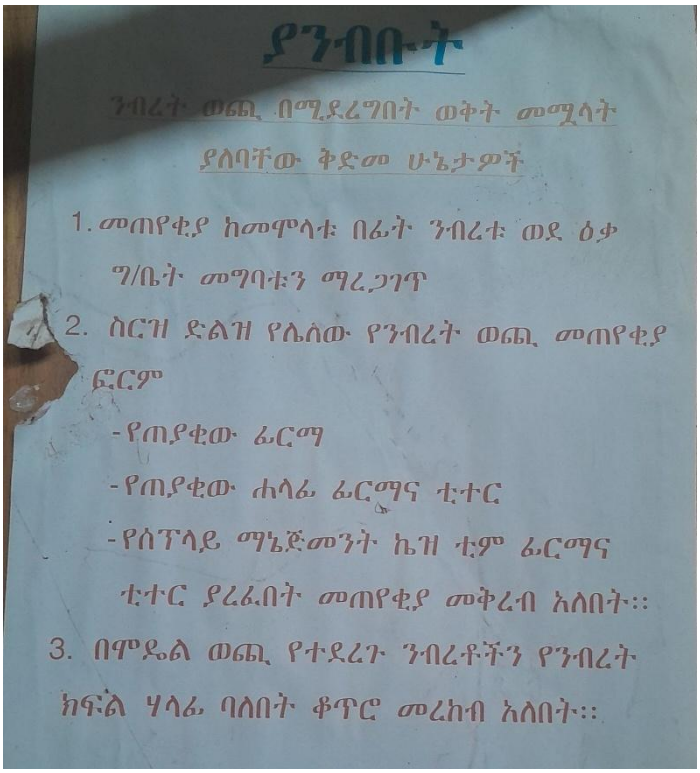
Megersa Daraje Abatu
Head,Computer Science & Engineering Department

CC

• CSE Department

The above letter is a **formal cooperation / permission letter** from **ASTU**, issued by the **CSE Department**, requesting different university offices to **support students who are conducting a Software Requirements Engineering (SRE) project**.

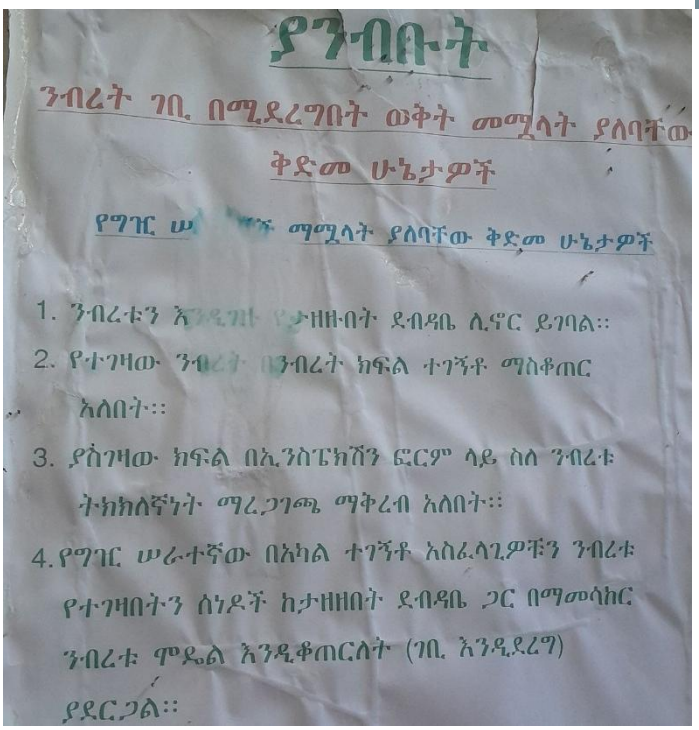
The procedures we follow when making item request



ያንብቡት

ንብረት ወጪ በሚደረግበት ወቅት መሟላት ያለባቸው ቅድመ ሁኔታዎች

1. መጠየቂያ ከመሞላቱ በፊት ንብረቱ ወደ ዕቃ ግ/ቤት መግባቱን ማረጋገጥ
2. ስርዝ ድልዝ የሌለው የንብረት ወጪ መጠየቂያ ፎርም
   - የጠያቂው ፊርማ
   - የጠያቂው ሐላፊ ፊርማና ቲተር
   - የሰፕላይ ማኔጅመንት ኬዝ ቲም ፊርማና ቲተር ያረፈበት መጠየቂያ መቅረብ አለበት፡፡
3. በሞዴል ወጪ የተደረጉ ንብረቶችን የንብረት ክፍል ሃላፊ ባለበት ቆጥሮ መረከብ አለበት፡፡



ያንብቡት

ንብረት ገቢ በሚደረግበት ወቅት መሟላት ያለባቸው ቅድመ ሁኔታዎች

የግዢ ሠ[illegible] ማሟላት ያለባቸው ቅድመ ሁኔታዎች

1. ንብረቱን እ[illegible] ታዘዙበት ደብዳቤ ሊኖር ይገባል፡፡
2. የተገዛው ንብ[illegible]ንብረት ክፍል ተገኝቶ ማስቆጠር አለበት፡፡
3. ያስገዛው ክፍል በኢንስፔክሽን ፎርም ላይ ስለ ንብረቱ ትክክለኛነት ማረጋገጫ ማቅረብ አለበት፡፡
4. የግዢ ሠራተኛው በአካል ተገኝቶ አስፈላጊዎቹን ንብረቱ የተገዛበትን ሰነዶች ከታዘዘበት ደብዳቤ ጋር በማመሳከር ንብረቱ ሞዴል እንዲቆጠርለት (ገቢ እንዲደረግ) ያደርጋል፡፡

The procedures we follow when transferring the item

The above pictures are locate in central store that announce the rule and regulation of the store we observe in the time of interviewing the warehouse managers.

APPENDIX B. SAMPLE INTERVIEW QUESTIONS IN REQUIREMENT GATHERING STAGE

A. Questions for Main Inventory Store Officer

1. How do you currently record items coming into and going out of the main store?
2. What documents do you use (paper forms, letters, notebooks)?
3. What information do you track about each item (quantity, supplier, date, serial number)?
4. What challenges do you face in the manual system?
5. How often do you face issues like lost records or wrong stock counts?
6. How do departments inform you when they take an item from their department store?

B. Questions for Department Storekeeper

1. What items do you manage in the department store?
2. How do you receive items from the main store?
3. How do you issue items to instructors or offices?
4. What happens after the department head approves a request?
5. Do you contact the main store after issuing an item?
6. What delays or errors happen in the paper-based process?
7. Do instructors complain about slow processes?
8. Would it help if you could see the main store stock in real time?

D. Questions for Instructors / Lecturers

1. How do you currently request an item?
2. How long does approval usually take?
3. Do you face delays in getting items?
4. Have you ever had a request lost or delayed?
5. Is it difficult to know which items are available?
6. Would you like to submit requests online?
7. Would tracking the status of your request be useful?
8. Do you want to see available stock before requesting?

APPENDIX C: SAMPLE FORMS

986
884
1026

The Federal Democratic Republic of Ethiopia
Adama Science and Technology University
Form Model 20 Report

Source Sub-Inventory : Main Store
Prepared By : SELAMDW
Destination : ADSTU

Model 20 Number : 0000884
Date Requested : 26-NOV-25
Transaction Type : Move Order Issue

S/No.	Item Code	Item Description	UOM	Quantity	Date Required	Remark
1	104101101.0047	Drink Water [illegible]	LT	62	26-NOV-25	
2	101101101.0006	Blue Pen	EA	60	26-NOV-25	
3	101101101.0005	Black Pen	EA	12	26-NOV-25	
4	101101101.0322	Transparent [illegible]	PM	5	26-NOV-25	
5	101102101.0049	Cloth Soap 20 [illegible]	EA	192	26-NOV-25	
6	101102101.0172	Toilet Soap 150gm	EA	144	26-NOV-25	

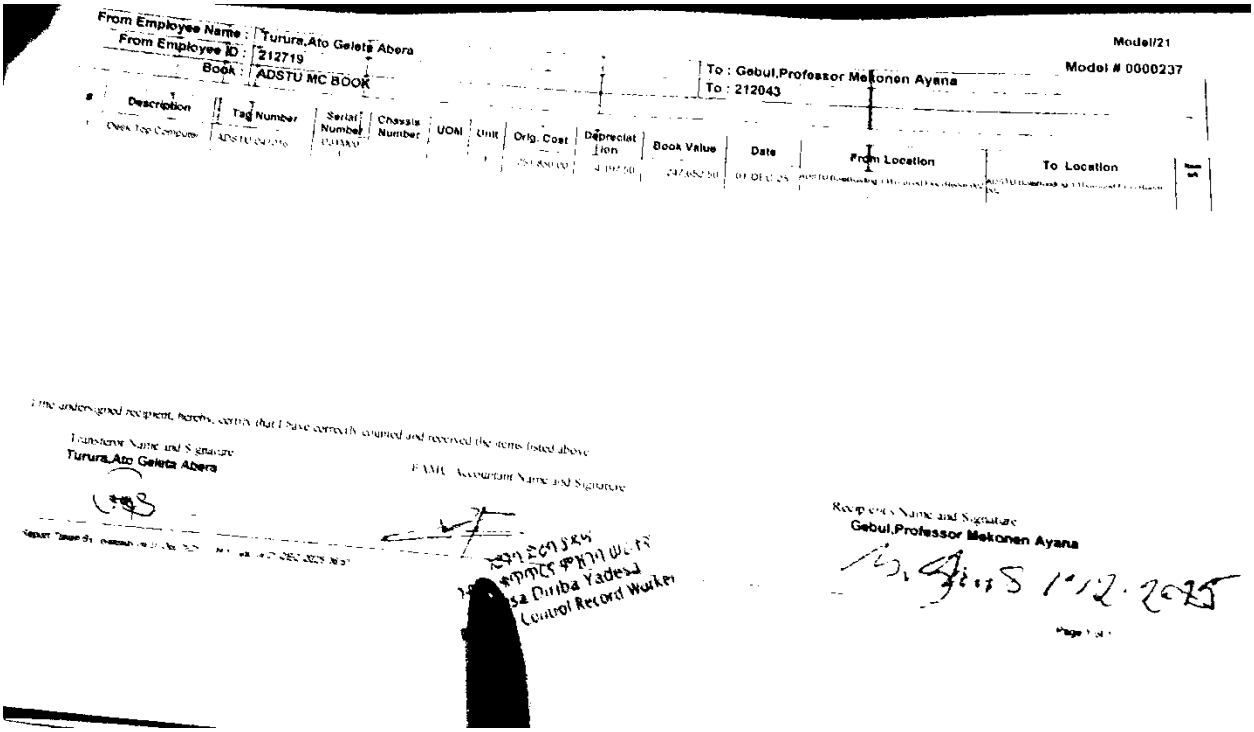
Requested By : Name : Wedajo Wiro Selam Desu

Signature

Approved By : Name :-

Signature

The above form is Model 20 form that is currently used for item request that is approved by the department head in manual system.



From Employee Name : Turura,Ato Geleta Abera
From Employee ID : 212719
Book : ADSTU MC BOOK

Model/21
Model # 0000237

To : Gebul,Professor Mekonen Ayana
To : 212043

#	Description	Tag Number	Serial Number	Chassis Number	UOM	Unit	Orig. Cost	Depreciation	Book Value	Date	From Location	To Location	Remarks
1	Desk Top Computer	ADSTU/24/076	[illegible]				[illegible]	[illegible]	[illegible]	01-DEC-25	[illegible]	[illegible]	

The undersigned recipient, hereby, certify that I have correctly counted and received the items listed above

Transferor Name and Signature
Turura,Ato Geleta Abera

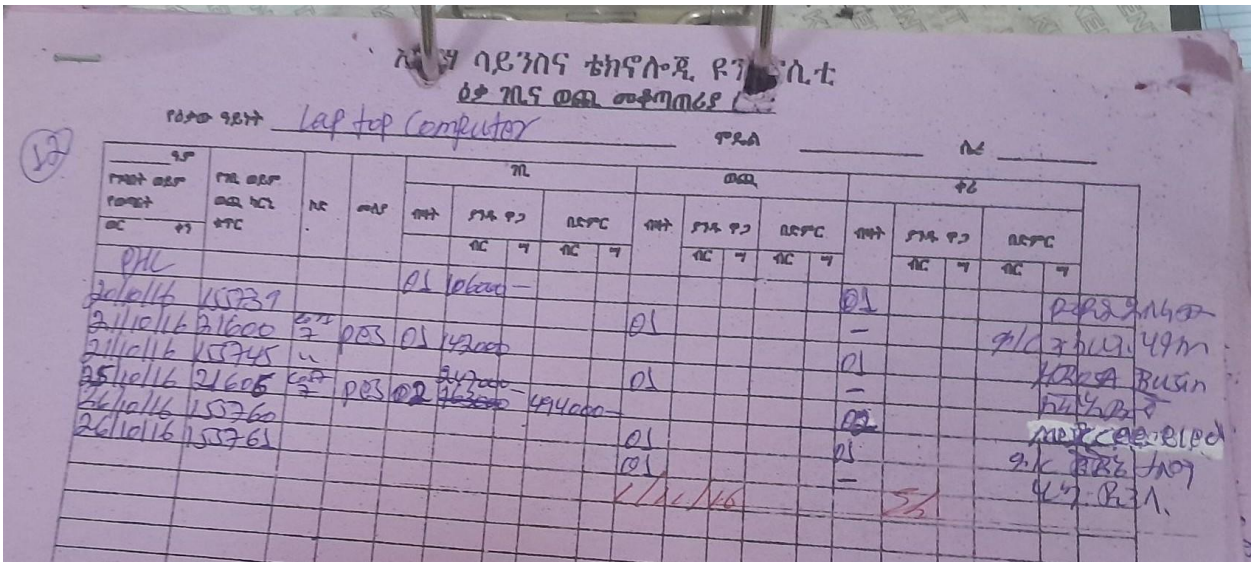
FAMC Accountant Name and Signature

Control Record Worker

Receiver's Name and Signature
Gebul,Professor Mekonen Ayana

1/12/2025

This is the form that the requester get the issued item and record in the above form.

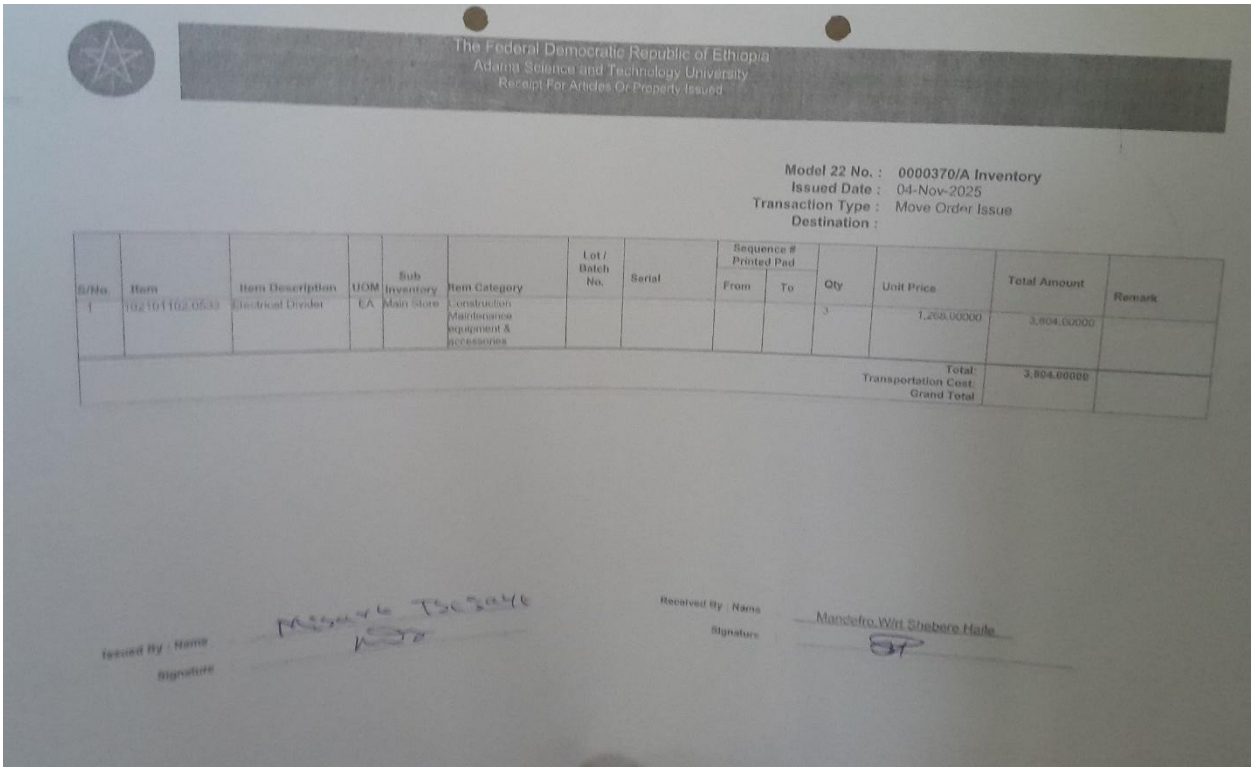


ዕቃ ገቢና ወጪ መቆጣጠሪያ

የዕቃው ዓይነት Laptop Computer ሞዴል ____ ቁጥር ____

ቀን	የገቢ ወይም የወጪ ቁጥር	ከየት	መለያ	ገቢ ብዛት	ገቢ ያገሉ ዋጋ ብር	ገቢ ያገሉ ዋጋ ሣ	ገቢ በድምር ብር	ገቢ በድምር ሣ	ወጪ ብዛት	ወጪ ያገሉ ዋጋ ብር	ወጪ ያገሉ ዋጋ ሣ	ወጪ በድምር ብር	ወጪ በድምር ሣ	ተሪ ብዛት	ተሪ ያገሉ ዋጋ ብር	ተሪ ያገሉ ዋጋ ሣ	ተሪ በድምር ብር	ተሪ በድምር ሣ	
PHC				01	10600	—													
20/10/16	155739								01					01					
21/10/16	21600		pos	01	42000									—					
21/10/16	155745								01					01					
25/10/16	21605		pos	02	94000	49400								—					
26/10/16	155760								01					01					
26/10/16	155761								01					—					
									1/11/16						52				

This is the item information quality, amount and birr. And also we get the number of receipt which the item is out from the store.



The Federal Democratic Republic of Ethiopia
Adama Science and Technology University
Receipt For Articles Or Property Issued

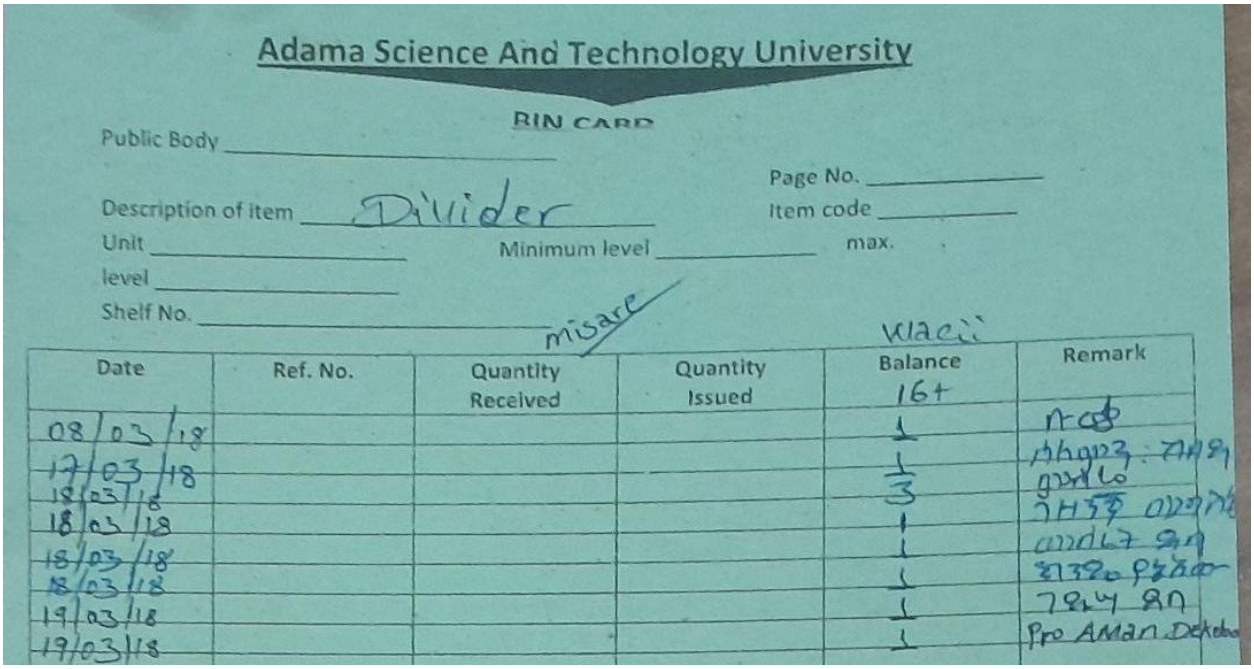
Model 22 No. : 0000370/A Inventory
Issued Date : 04-Nov-2025
Transaction Type : Move Order Issue
Destination :

S/No.	Item	Item Description	UOM	Sub Inventory	Item Category	Lot / Batch No.	Serial	Sequence # Printed Pad From	Sequence # Printed Pad To	Qty	Unit Price	Total Amount	Remark
1	102101102.0533	Electrical Divider	EA	Main Store	Construction Maintenance equipment & accessories					3	1,268.00000	3,804.00000	
											Total:	3,804.00000	
											Transportation Cost:		
											Grand Total		

Issued By : Name
Signature

Received By : Name Mandefro Wirt Shebere Haile
Signature

Model 22 form- for transferring purpose from one user to the other.



Adama Science And Technology University

BIN CARD

Public Body _____

Page No. _____

Description of item Divider

Item code _____

Unit _____

Minimum level _____ max.

level _____

Shelf No. _____

Date	Ref. No.	Quantity Received	Quantity Issued	Balance	Remark
08/03/18				1	
17/03/18				1/3	
18/03/18				1	
18/03/18				1	
18/03/18				1	
18/03/18				1	
19/03/18				1	
19/03/18				1	

It tells the amount of item stored in the store and decreased when it is out from the store in paper form.